Naveen Gupta

ngupta.maths@gmail.com +91-9664060887
Homepage

Education

October, 2025 – Present
Postdoctoral Scholar
Department of Statistics
The Pennsylvania State University, USA.
Advisor: Prof. Bharath K Sriperumbudur

July, 2025 – September 2025
Research Associate
Department of Mathematics
Indian Institute of Technology Delhi, New Delhi.
Advisor: Prof. S. Sivananthan

October, 2020 – July 2025
Ph.D.
Department of Mathematics
Indian Institute of Technology Delhi, New Delhi.
Thesis title: *Convergence Analysis of Learning Algorithms for Functional Data*
Supervisor: Prof. S. Sivananthan

July, 2018 – June, 2020
Master of Science
Department of Mathematics and Statistics
Indian Institute of Technology Delhi, New Delhi.
Project title: *Construction of Gabor Frame from the Translation and Modulation of a L^2 function*
Supervisor: Prof. S. Sivananthan

July, 2015 – June, 2018
Bachelor of Science (Hons.)
Department of Mathematics
Rajrishi Govt. college.
Affiliated to the Rajrishi Bhartrihari Matsya University.

Research interest

Learning Theory, Kernel Methods, Functional Data Analysis
My research focuses on functional data analysis, statistical learning theory, and reproducing kernel Hilbert spaces (RKHS), with particular emphasis on functional linear and nonlinear regression models. I work on theoretical and computational aspects of infinite-dimensional learning problems, including regularization methods, covariance and integral operators, kernel-based estimation, and minimax convergence analysis.

Publication

Journal Publication

1 **N. Gupta** and S. Sivananthan, "Convergence analysis of function-on-function polynomial regression model," *Analysis and Applications*, 2026. DOI: 10.1142/S0219530526500727.

2 **N. Gupta** and S. Sivananthan, "Convergence analysis of regularised nyström method for functional linear regression," *Inverse Problems*, 2025. DOI: 10.1088/1361-6420/adbfb6.

3 **N. Gupta** and S. Sivananthan, "Revisiting general source condition in learning over a Hilbert space," *Inverse Problems*, 2025. DOI: 10.1088/1361-6420/adec15.

4 **N. Gupta**, S. Sivananthan, and B. K. Sriperumbudur, "Optimal rates for functional linear regression with general regularization," *Appl. Comput. Harmon. Anal.*, vol. 76, Paper No. 101745, 29, 2025, ISSN: 1063-5203,1096-603X. DOI: 10.1016/j.acha.2024.101745.

5 **N. Gupta**, S. Sivananthan, and B. K Sriperumbudur, "Convergence analysis of kernel conjugate gradient for functional linear regression," *Journal of applied and numerical analysis*, 2023. URL: http://jnam.lnu.edu.ua/pdf/y2023v01/y2023(1)03_Gupta_Sivananthan_Sriperumbudur.pdf.

Preprint

1 **N. Gupta** and B. K Sriperumbudur, *Minimax optimal estimation of mean and covariance functions with spectral regularization.*

2 **N. Gupta**, V. Silmana, and S. Sivananthan, *Scalable operator learning via nystr om approximation with denoising applications.*

Advance Study

Operator Theory
Algebraic Topology
Fourier Analysis
Mathematical Analysis in Learning Theory
Abstract Harmonic Analysis

C. Programming
Matlab
Python (Basic)

Teaching Experience

October, 2020 – July, 2025

Teaching Assistant, IIT Delhi
MTL100: Calculus
MTL101: Linear Algebra and Differential Equation
MTL122: Real and Complex Analysis
MTL506: Complex Analysis
MTL510: Measure and Integration

Workshops and Conferences

1. Presented my work "**Convergence Analysis of Kernel Conjugate Gradient for Functional Linear Regression**" in "**Maths4DL Conference on Inverse Problems and Deep Learning**" held at University of Bath from 07th July 2025 to 09th July 2025.
2. Attended the workshop "**AIS- Statistical Learning Theory, 2025**" held at IISER Kolkata from 12th May 2025 to 24th May 2025.
3. Presented my work "**Optimal rates for functional linear regression with general regularization**" in "**Symposium on Sparsity and Singular Structures 2024**" held at RWTH Aachen University from 26th February 2024 to 29th February 2024.
4. Presented my work "**Convergence Analysis of Kernel Conjugate Gradient for Functional Linear Regression**" in the conference "**International Conference on Spectral and Approximation Theory (ICSAT-2023)**" held at Cochin University of Science And Technology, Kerala from 27th November 2023 to 30th November 2023.
5. Attended "**International Conference on Analysis, Inverse Problems and Applications-2022 (ICAIPA-2022)**" held at Indian Institute of Technology Madras from 18th July 2022 to 21th July 2022.

6. Attended "**Focus Program on Analytic Function Spaces and their Applications**" organised by The Fields Institute for Research in Mathematical Sciences from 1st July 2021 to 31st December 2021 (Virtual).